

Learning Bimanual Dexterous Jar Opening with Finger Gaitting

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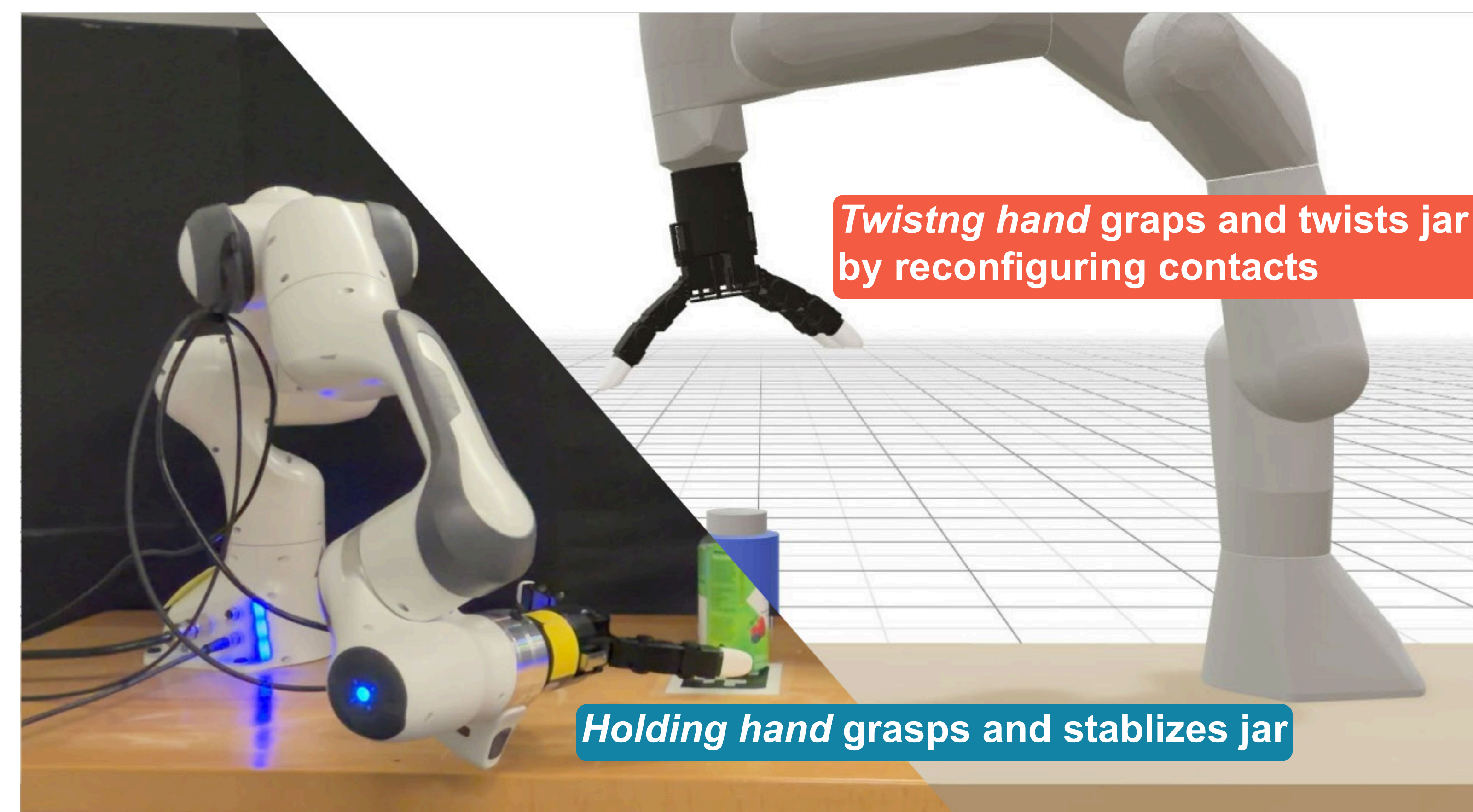
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One RL-trained policy grasps and twists open jars with a bimanual dexterous system.

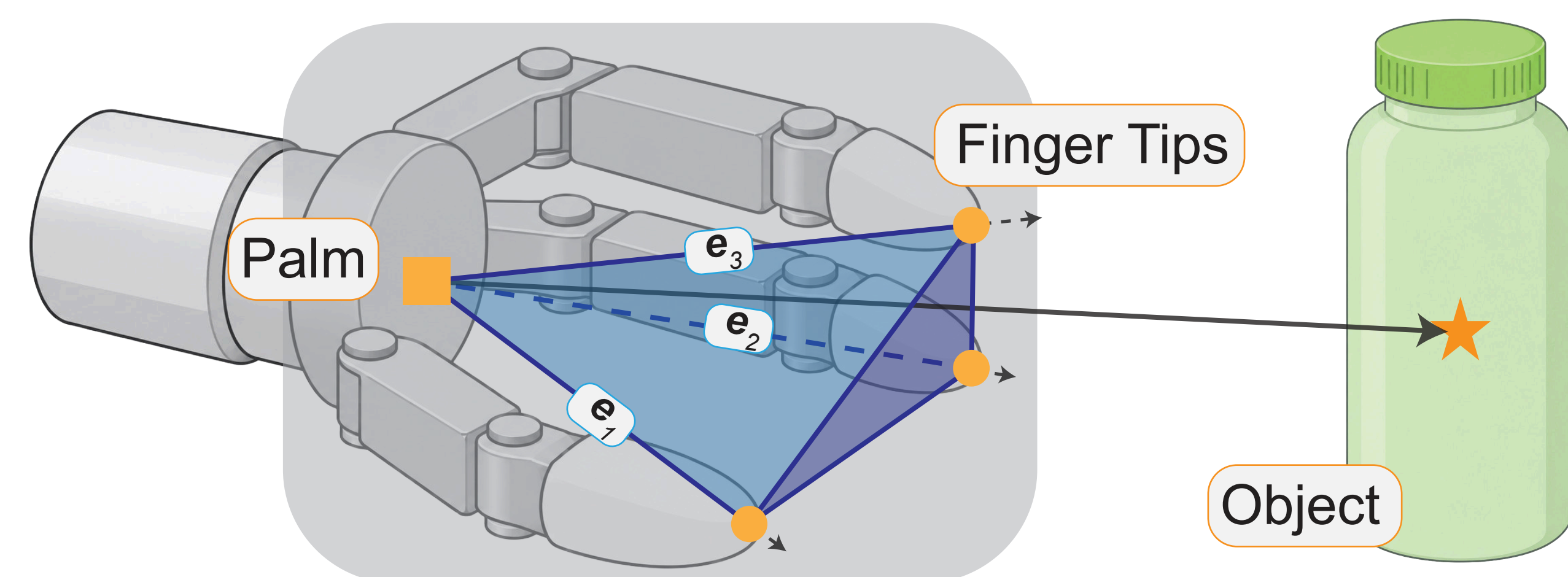
One hand reconfigures finger-object contacts to enable sustained lid rotation while the other hand holds the jar.

Task and Method

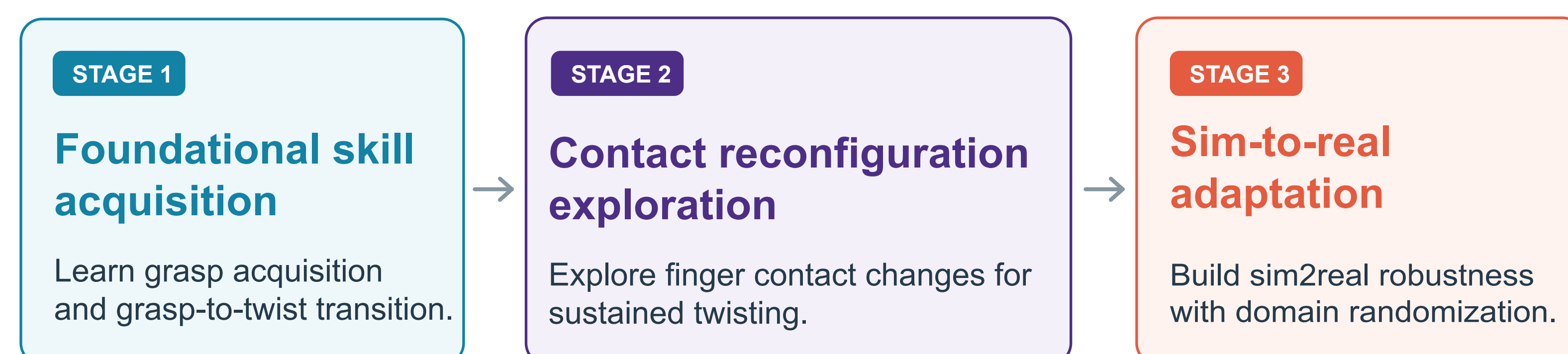


Physical system (left) and simulation (right)

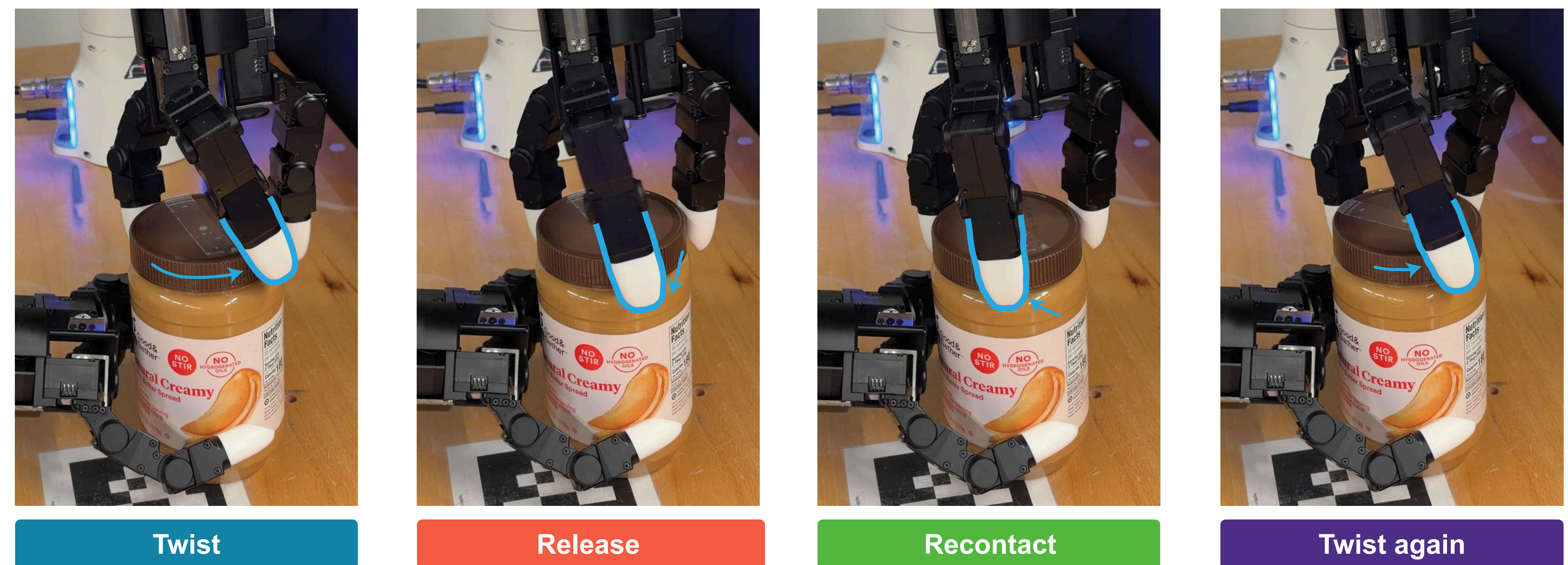
We guide fingers to spatial arrangement during grasping and signal the grasp-to-twist transition.



We train the policy with a three-stage curriculum.



Finger Gaitting in Real World



Fingers release, reposition, and re-establish contact to continue twisting beyond the motion range of a fixed grasp.

Physical evaluation on household containers



88% Six household containers
10 trials per container
53 / 60 trials

Success: complete lid thread disengagement, verified by open-loop lid lifting by twisting hand.



Project Page
scan for details

